Vehicle Parking Management System

Branch Coverage Report

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Executed 8 November 2026 on commit 39e0e56. 748 of 748 Jest tests passed in 39 suites.

1. Scope and method

Branch coverage counts each way a decision can go: both sides of an if or ?:, each operand of || and &&, each case, and whether a default parameter value is used. It was measured with Istanbul during the full Jest run (npm run test:coverage) over server/src, excluding the start-up file index.js.

2. Summary

Metric	Covered	Total	Coverage
Branches	626	675	92.74%

49 of 71 files have 100% branch coverage. All business rules (slot assignment, fees, payments, reservations, roles, lockout, session timeout) have both sides of every decision tested; the 49 branches not taken are explained in Section 4.

3. Branch coverage by file

File (server/src/)	Covered	Total	Branch coverage
app.js	1	2	50.00%
config.js	13	22	59.09%
db.js	0	0	100%
errors.js	2	2	100%
middleware/authenticate.js	2	2	100%
middleware/errorHandler.js	15	17	88.23%
middleware/requireRole.js	10	10	100%
modules/admin/admin.controller.js	0	0	100%
modules/admin/admin.repository.js	8	8	100%
modules/admin/admin.routes.js	0	0	100%
modules/admin/admin.service.js	15	18	83.33%
modules/admin/admin.validation.js	23	26	88.46%
modules/audit/audit.controller.js	0	0	100%
modules/audit/audit.repository.js	4	4	100%
modules/audit/audit.routes.js	0	0	100%
modules/audit/audit.service.js	0	0	100%
modules/audit/audit.validation.js	23	25	92.00%
modules/audit/auditActions.js	0	0	100%
modules/history/history.controller.js	0	0	100%
modules/history/history.repository.js	0	0	100%
modules/history/history.routes.js	0	0	100%
modules/history/history.service.js	2	2	100%
modules/history/history.validation.js	19	19	100%
modules/parking/parking.controller.js	9	12	75.00%
modules/parking/parking.repository.js	24	27	88.88%
modules/parking/parking.routes.js	0	0	100%
modules/parking/parking.service.js	42	42	100%
modules/parking/parking.validation.js	33	35	94.28%

File (server/src/)	Covered	Total	Branch coverage
modules/parking/sessionStatuses.js	0	0	100%
modules/payments/payments.controller.js	1	2	50.00%
modules/payments/payments.repository.js	2	2	100%
modules/payments/payments.routes.js	0	0	100%
modules/payments/payments.service.js	29	29	100%
modules/payments/payments.validation.js	54	55	98.18%
modules/rates/fees.js	4	4	100%
modules/rates/rates.controller.js	0	0	100%
modules/rates/rates.repository.js	3	3	100%
modules/rates/rates.routes.js	0	0	100%
modules/rates/rates.service.js	2	2	100%
modules/rates/rates.validation.js	26	27	96.29%
modules/reports/reports.controller.js	0	0	100%
modules/reports/reports.repository.js	2	2	100%
modules/reports/reports.routes.js	0	0	100%
modules/reports/reports.service.js	10	10	100%
modules/reports/reports.validation.js	23	24	95.83%
modules/reservations/reservations.controller.js	0	0	100%
modules/reservations/reservations.repository.js	3	4	75.00%
modules/reservations/reservations.routes.js	0	0	100%
modules/reservations/reservations.service.js	14	15	93.33%
modules/reservations/reservations.validation.js	33	33	100%
modules/slots/slotStatuses.js	0	0	100%
modules/slots/slotTypes.js	0	0	100%
modules/slots/slots.controller.js	6	6	100%
modules/slots/slots.repository.js	2	2	100%
modules/slots/slots.routes.js	0	0	100%
modules/slots/slots.service.js	7	11	63.63%
modules/slots/slots.validation.js	37	39	94.87%
modules/users/password.js	2	2	100%
modules/users/roles.js	0	0	100%
modules/users/sessions.repository.js	2	2	100%
modules/users/users.controller.js	0	0	100%
modules/users/users.repository.js	8	9	88.88%
modules/users/users.routes.js	0	0	100%
modules/users/users.service.js	33	36	91.66%
modules/users/users.validation.js	33	34	97.05%
modules/vehicles/vehicles.controller.js	0	0	100%
modules/vehicles/vehicles.repository.js	2	2	100%
modules/vehicles/vehicles.routes.js	0	0	100%
modules/vehicles/vehicles.service.js	9	12	75.00%
modules/vehicles/vehicles.validation.js	27	28	96.42%
sessionCookie.js	7	7	100%
Total	**626**	**675**	**92.74%**

4. Every branch not taken, and why

Location	Kind	Code	Category	Reason not tested
app.js:69	if	if (fs.existsSync(CLIENT_DIST)) {	Environment	The built client (client/dist) is served only when it exists. Jest runs without a build; the browser tests cover the served client.

Location	Kind	Code	Category	Reason not tested
config.js:8	if	if (fs.existsSync(envFile)) {	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
config.js:11	if	if (process.env[key] === undefined) process.env[key] = value;	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
config.js:15	binary-expr	const env = process.env.NODE_ENV || 'development';	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
config.js:20	binary-expr	port: Number(process.env.PORT) || 4000,	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
config.js:22	binary-expr	host: process.env.DB_HOST || '127.0.0.1',	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
config.js:23	binary-expr	port: Number(process.env.DB_PORT) || 3306,	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
config.js:24	binary-expr	user: process.env.DB_USER || 'parking_app',	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
config.js:25	binary-expr	password: process.env.DB_PASSWORD || '',	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
config.js:26	binary-expr	database: process.env.DB_NAME || 'parking_dev',	Environment	Default used when an environment variable is missing. Tests always set NODE_ENV, DB_* and PORT, and the .env file exists.
middleware/errorHandler.js:6	if	if (res.headersSent) return next(err);	Defensive guard	An error after the response has started. No route streams a response, so this cannot happen today.
middleware/errorHandler.js:26	binary-expr	console.error(`${err && err.code ? `[${err.code}] ` : ''}${(err && err.stack) || err}`);	Defensive guard	Logging fallback for a thrown value that has no stack, e.g. `throw 'text'`. Server code only throws Error objects.
modules/admin/admin.service.js:20	if	if (err.code === 'ER_DUP_ENTRY') {	Unexpected error	Re-throws a database error other than the one handled, so it reaches the 500 handler. Needs a database fault; the 500 path itself is tested in reliability.test.js.
modules/admin/admin.service.js:42	if	if (err.code === 'ER_ROW_IS_REFERENCED_2') {	Unexpected error	Re-throws a database error other than the one handled, so it reaches the 500 handler. Needs a database fault; the 500 path itself is tested in reliability.test.js.
modules/admin/admin.service.js:68	if	if (err.code === 'ER_DUP_ENTRY') {	Unexpected error	Re-throws a database error other than the one handled, so it reaches the 500 handler. Needs a database fault; the 500 path itself is tested in reliability.test.js.
modules/admin/admin.validation.js:12	default-arg	function validateUserFilter(query = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/admin/admin.validation.js:26	default-arg	function validateNewUser(body = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/admin/admin.validation.js:32	if	if (!err.fields) throw err;	Unexpected error	Re-throws an error that is not a validation error. The validator only throws validation errors.
modules/audit/audit.validation.js:9	cond-expr	return typeof value === 'string' && /^\d{1,6}$/.test(value) ? Number(value) : NaN;	Defensive guard	A query value that is not a string (an array from ?page=1&page=2). Express parses the tested queries as strings.
modules/audit/audit.validation.js:13	default-arg	function validateAuditQuery(query = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/parking/parking.controller.js:27	binary-expr	const registrationNumber = cleanRegistration((req.body || {}).registrationNumber);	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/parking/parking.controller.js:38	binary-expr	const input = validateExit(req.body || {});	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/parking/parking.controller.js:50	binary-expr	const session = await service.moveVehicle(validateMove(req.body || {}));	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.

Location	Kind	Code	Category	Reason not tested
modules/parking/parking.repository.js:4	default-arg	async function findVehicleById(vehicleId, userId = null) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/parking/parking.repository.js:98	if	if (candidates.length < batch) break; // nothing more to try	Concurrency	Every slot in a full candidate batch was taken by another transaction at the same moment, so a larger batch is tried. Needs more simultaneous entries than free slots in the first batch; the 100-user test did not reach it.
modules/parking/parking.repository.js:167	if	if (rows[0]) return rows[0];	Concurrency	A reserved candidate slot was locked by another transaction between the two queries. Timing-dependent.
modules/parking/parking.validation.js:48	default-arg	function validateExit(body = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/parking/parking.validation.js:65	default-arg	function validateMove(body = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/payments/payments.controller.js:10	binary-expr	const input = validatePayment(req.body || {});	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/payments/payments.validation.js:67	default-arg	function validatePayment(body = {}, now = new Date()) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/rates/rates.validation.js:24	default-arg	function validateRate(body = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/reports/reports.validation.js:19	default-arg	function validateRange(query = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/reservations/reservations.repository.js:29	binary-expr	return rows[0] || null;	Defensive guard	Lookup that finds nothing. The service checks existence first, so callers never ask for a missing row.
modules/reservations/reservations.service.js:40	if	if (	Defensive guard	Second check of the reservation times. validateReservation already refuses bad times, so the service never receives them.
modules/slots/slots.service.js:11	default-arg	async function listWithCounts({ status = null, type = null } = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/slots/slots.service.js:11	default-arg	async function listWithCounts({ status = null, type = null } = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/slots/slots.service.js:11	default-arg	async function listWithCounts({ status = null, type = null } = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/slots/slots.service.js:28	if	if (err.code === 'ER_DUP_ENTRY') {	Unexpected error	Re-throws a database error other than the one handled, so it reaches the 500 handler. Needs a database fault; the 500 path itself is tested in reliability.test.js.
modules/slots/slots.validation.js:18	default-arg	function validateNewSlot(body = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/slots/slots.validation.js:36	default-arg	function validateSlotUpdate(body = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/users/users.repository.js:51	binary-expr	const row = rows[0] || {};	Defensive guard	A COUNT query always returns one row, so the empty fallback is never used.
modules/users/users.service.js:75	default-arg	async function getUserForToken(token, { keepAlive = true } = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/users/users.service.js:75	default-arg	async function getUserForToken(token, { keepAlive = true } = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.
modules/users/users.service.js:86	if	if (!user) return null;	Defensive guard	A session whose user no longer exists. Removing a user deletes their sessions (foreign key cascade), so this cannot happen.
modules/users/users.validation.js:45	default-arg	function validateLogin(body = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.

Location	Kind	Code	Category	Reason not tested
modules/vehicles/vehicles.service.js:7	if	if (err.code === 'ER_DUP_ENTRY') {	Unexpected error	Re-throws a database error other than the one handled, so it reaches the 500 handler. Needs a database fault; the 500 path itself is tested in reliability.test.js.
modules/vehicles/vehicles.service.js:52	if	if (err instanceof HttpError) throw err;	Unexpected error	Re-throws a database error other than the one handled, so it reaches the 500 handler. Needs a database fault; the 500 path itself is tested in reliability.test.js.
modules/vehicles/vehicles.service.js:63	if	if (err.code === 'ER_ROW_IS_REFERENCED_2') {	Unexpected error	Re-throws a database error other than the one handled, so it reaches the 500 handler. Needs a database fault; the 500 path itself is tested in reliability.test.js.
modules/vehicles/vehicles.validation.js:25	default-arg	function validateVehicle(body = {}) {	Defensive default	Default for a missing argument or request body. Every caller passes one, and express.json() always sets req.body.

By category

Category	Branches	Meaning
Defensive default	22	Default for a missing argument or body that every caller supplies.
Unexpected error	8	Re-throw of a database error other than the expected one; would need fault injection.
Environment	10	Fallbacks used only when configuration is missing.
Defensive guard	7	Checks for states other code or database constraints already prevent.
Concurrency	2	Paths taken only under specific timing between simultaneous transactions.
Total	**49**	

5. Branches found and closed during the testing week

The first coverage run (743 tests, 91.85% branches) showed five behaviours with no test, behind branches that were never taken. Tests were added for each, raising coverage to the figure above:

- Removing an attendant who recorded payments is refused (admin.service.js).
- Vehicle model and colour length limits (vehicles.validation.js).
- Luhn check when a doubled digit is over 9 (payments.validation.js).
- Impossible date with a time zone in a reservation (reservations.validation.js).
- Lockout message with under a minute left says "1 minute" (users.service.js).